

Package ‘dynTools’

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Contents

CheckDynData	2
CombineByIDTime	4
DeleteInitialNA	5
DeleteObservedAllNA	7
DeltaTByID	8
DetrendByID	10
DiagnoseScaleByID	12
DiagnosticsByID	13
DropByID	15
ElapsedTimeByID	16

ElapsedTimeByIDCT	18
FilterByID	21
FilterObservedRows	22
FlagDiagnosticsByID	24
GetDropID	26
GetObservedInitial	27
InitialNA	29
InsertNA	30
LagByID	31
MakeClockTime	33
PlotByID	34
RegularizeTimeByID	37
ReplaceMissingCode	39
ResolveCloseTimeByID	40
ResolveDuplicateIDTime	42
RoundClockTime	44
ScaleByID	45
ScreenByID	47
SubsetByID	49
SummaryByID	51
TrimInitialRowsByID	52
Index	55

CheckDynData

Check Dynamic Modeling Data

Description

The function checks whether a data frame has the variables and structure required by the dynamic modeling utility functions.

Usage

```

CheckDynData(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  require_unique = TRUE,
  require_numeric_time = TRUE,
  require_numeric_observed = TRUE,
  require_numeric_covariates = FALSE,
  min_rows = 2L
)

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.
<code>require_unique</code>	Logical. If TRUE, require unique id-time combinations.
<code>require_numeric_time</code>	Logical. If TRUE, require the time variable to be numeric.
<code>require_numeric_observed</code>	Logical. If TRUE, require observed variables to be numeric.
<code>require_numeric_covariates</code>	Logical. If TRUE, require covariates to be numeric.
<code>min_rows</code>	Positive integer. Minimum number of rows required per ID.

Value

Returns TRUE invisibly if all checks pass.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = rnorm(6),
  y2 = rnorm(6)
)
CheckDynData(
```

```

data = data,
id = "id",
time = "time",
observed = c("y1", "y2")
)

```

CombineByIDTime

Combine Data Sets by ID and Time

Description

The function appends observed variables from one data set to another data set using exact id-time matches. Rows in data define the output id-time structure. If an id-time row in data is not present in append_data, missing values are inserted for the appended observed variables.

Usage

```
CombineByIDTime(data, append_data, id, time, observed, covariates = NULL)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
append_data	Data frame. A data frame object containing the id and time columns and the observed variables to append to data.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  cov = c(10, 10, 10, 20, 20, 20)
)

append_data <- data.frame(
  id = c(1, 1, 2),
  time = c(1, 3, 2),
  y1 = c(11, 13, 22),
  y2 = c(101, 103, 202)
)

CombineByIDTime(
  data = data,
  append_data = append_data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  covariates = "cov"
)
```

DeleteInitialNA

*Delete Initial Rows With Missing Observed Values by ID***Description**

The function removes leading rows by ID when the observed variables contain missing values. This process is repeated until the first row per ID no longer has missing observed values. Covariates are retained when selecting and sorting the data, but they are not used to determine whether an initial row should be removed.

Usage

```
DeleteInitialNA(data, id, time, observed, covariates = NULL)
```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.

Details

This is a strict helper because the first retained row must be complete on all observed variables. For more flexible trimming, prefer [TrimInitialRowsByID\(\)](#) with `min_nonmissing`.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, 10, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, 10, 11, 12, 13, 14),
  y3 = c(1, NA, 3, 4, 5, 10, 11, 12, 13, 14)
)
data

DeleteInitialNA(
  data = data,
  id = "id",
```

```
time = "time",  
observed = paste0("y", 1:3)  
)
```

DeleteObservedAllNA	<i>Delete Rows With All Observed Variables Missing</i>
---------------------	--

Description

The function removes rows where all observed variables are missing.

Usage

```
DeleteObservedAllNA(data, id, time, observed, covariates = NULL)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

Details

Covariates are retained in the returned data, but they are not used to decide whether a row should be deleted.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, NA, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, NA, 11, 12, 13, 14),
  y3 = c(NA, NA, 3, 4, 5, NA, 11, 12, 13, 14)
)
data

DeleteObservedAllNA(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3)
)
```

DeltaTByID

Summarize Time Intervals by ID

Description

The function returns a diagnostic table describing the spacing of the time variable within each ID.

Usage

```
DeltaTByID(
  data,
  id,
  time,
  observed = NULL,
  covariates = NULL,
  tolerance = sqrt(.Machine$double.eps)
)
```


Arguments

<code>data</code>	Data frame. A data frame object.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. Optional vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. Optional vector of character strings of the names of the covariates in the data.
<code>tolerance</code>	Numeric. Tolerance used to determine whether the time intervals are regular.

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = c(1, 2, 3, 1, 3, 4),
  y = rnorm(6)
)
data

DeltaTByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y"
)
```

DetrendByID

Detrend Observed Variables by ID

Description

The function removes polynomial time trends from observed variables within each ID. Missing observed values are ignored when estimating the trend and remain missing in the detrended variables.

Usage

```
DetrendByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  degree = 1L,
  replace = FALSE,
  keep_mean = TRUE,
  prefix = "detrend",
  warn_skipped = TRUE,
  drop_skipped_ids = TRUE
)
```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.
<code>degree</code>	Non-negative integer. Degree of the polynomial trend. Use <code>degree = 0</code> for mean centering by ID when <code>keep_mean = FALSE</code> . If <code>degree = 0</code> and <code>keep_mean = TRUE</code> , the original observed values are returned.
<code>replace</code>	Logical. If <code>TRUE</code> , replace observed variables with detrended variables. If <code>FALSE</code> , append detrended variables to the data frame.
<code>keep_mean</code>	Logical. If <code>TRUE</code> , preserve the original within-ID mean after detrending. If <code>FALSE</code> , return ordinary residuals from the within-ID trend model.
<code>prefix</code>	Character string. Prefix for detrended variables when <code>replace = FALSE</code> .

`warn_skipped` Logical. If TRUE, warn when one or more ID-variable combinations cannot be detrended because of too few usable observations, too few unique time values, or a rank-deficient trend design.

`drop_skipped_ids` Logical. If TRUE, remove all rows for IDs with at least one observed variable that cannot be detrended.

Details

If `keep_mean = TRUE`, the within-ID observed mean is added back after removing the fitted time trend. This removes temporal drift while preserving each ID's observed-variable level. If `keep_mean = FALSE`, the output is the ordinary residual from the within-ID polynomial trend model.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 4),
  time = rep(1:4, times = 2),
  y = rep(1:4, times = 2)
)
data
```

```
DetrendByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  degree = 1,
  keep_mean = TRUE
)
```

```
DetrendByID(
  data = data,
  id = "id",
```

```

time = "time",
observed = "y",
degree = 1,
keep_mean = FALSE
)

```

DiagnoseScaleByID

Diagnose Scaling by ID

Description

The function computes ID-variable-level diagnostics before within-ID scaling. It reports within-ID variability, missingness, non-finite values, and the largest standardized value that would be produced by scaling.

Usage

```

DiagnoseScaleByID(
  data,
  id,
  time,
  observed,
  sd_min = 0.1,
  z_cut = 6,
  min_n = 3L,
  flagged_only = FALSE
)

```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
sd_min	Numeric scalar or NULL. Minimum acceptable within-ID standard deviation before within-ID scaling. If NULL, low-SD flagging is skipped.
z_cut	Numeric scalar. Absolute standardized-value threshold used to flag potentially extreme values after within-ID scaling.
min_n	Positive integer. Minimum number of finite observations required within each ID-variable combination.
flagged_only	Logical. If TRUE, return only flagged ID-variable combinations. If FALSE, return all ID-variable combinations.

Value

Returns a data frame with one row per ID-variable combination.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(1, 2, 3, 4, 20, 2, 2, 2, 2, 2),
  y2 = c(5, 4, 3, 2, 1, 1, 1, 1, 1, 2)
)

DiagnoseScaleByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  z_cut = 3
)
```

Description

The function computes ID-level diagnostics for intensive longitudinal data. Diagnostics include the number of finite observed rows, number of complete rows, proportion of rows with no finite observed values, duplicate ID-time rows, time gaps, non-finite observed values, within-ID standard deviations, and counts of extreme observed values.

Usage

```

DiagnosticsByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  missing_codes = NULL,
  min_nonmissing = 1L,
  extreme_cut = c(4, 5, 6),
  sd_cut = c(0.1, 0.05),
  time_scale = 1,
  posix_unit = "hours"
)

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.
<code>missing_codes</code>	NULL or vector. Values to temporarily treat as missing when computing diagnostics. The input data are not modified.
<code>min_nonmissing</code>	Positive integer. Minimum number of finite observed variables required for a row to count as an observed row.
<code>extreme_cut</code>	Numeric vector. Absolute-value cutoffs used to count extreme values.
<code>sd_cut</code>	Numeric vector. Within-ID standard-deviation cutoffs used to count low-variance variables.
<code>time_scale</code>	Positive number. Multiplier applied to numeric time gaps. For example, use <code>time_scale = 24</code> when time is measured in days and gaps should be reported in hours.
<code>posix_unit</code>	Character string. Unit for gaps when time is a POSIXt or Date variable. Passed to <code>as.numeric.difftime()</code> .

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

DiagnosticsByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

DropByID

*Drop IDs From a Data Frame***Description**

The function removes rows belonging to one or more IDs. It is a small convenience wrapper used after ID-level diagnostics identify IDs for exclusion or sensitivity analysis.

Usage

```
DropByID(data, id, drop = NULL)
```

Arguments

data	Data frame.
id	Character string. Name of the ID variable.
drop	Vector. ID values to remove. If NULL or empty, data is returned unchanged.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2, 3, 3),
  y = 1:6
)

DropByID(
  data = data,
  id = "id",
  drop = c(2, 3)
)
```

ElapsedTimeByID

Create Elapsed Time by ID

Description

The function creates an elapsed-time variable within ID. For date-time variables, elapsed time is computed with `difftime()`. For numeric time variables, elapsed time is computed by subtraction, optionally converting from `input_units` to `units`.

Usage

```
ElapsedTimeByID(
  data,
  id,
  time,
  output = "time_elapsed",
  units = "days",
  origin = c("by_id", "global"),
```



```

    input_units = NULL,
    replace = FALSE
  )

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>output</code>	Character string. Name of the elapsed-time variable.
<code>units</code>	Character string. Units for the output elapsed time.
<code>origin</code>	Character string. If <code>origin = "by_id"</code> , each ID's origin is its own minimum non-missing time. If <code>origin = "global"</code> , all IDs use the global minimum non-missing time.
<code>input_units</code>	Character string or NULL. Units of numeric input time. If NULL, numeric time is assumed to already be in the desired output scale and only subtraction is performed.
<code>replace</code>	Logical. If TRUE, replace the original time variable. If FALSE, append output.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```

data <- data.frame(
  id = c(1, 1, 2, 2),
  datetime = as.POSIXct(
    c(
      "2020-01-01 00:00:00",
      "2020-01-02 00:00:00",

```

```

      "2020-01-03 12:00:00",
      "2020-01-04 00:00:00"
    ),
    tz = "UTC"
  )
)
data

ElapsedTimeByID(
  data = data,
  id = "id",
  time = "datetime",
  output = "time",
  units = "days"
)

```

ElapsedTimeByIDCT

Create CT-Scaled Elapsed Time by ID

Description

The function creates an elapsed-time variable within ID and rescales it for continuous-time state-space modeling. The rescaling divides elapsed time by a typical positive consecutive time interval so that the typical `delta_t` is approximately 1.

Usage

```

ElapsedTimeByIDCT(
  data,
  id,
  time,
  output = "time_ct",
  units = "hours",
  origin = c("by_id", "global"),
  input_units = NULL,
  scale = c("mean_dt", "median_dt"),
  scale_value = NULL,
  replace = FALSE
)

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.

time	Character string. A character string of the name of the TIME variable in the data.
output	Character string. Name of the elapsed-time variable.
units	Character string. Units for the output elapsed time.
origin	Character string. If origin = "by_id", each ID's origin is its own minimum non-missing time. If origin = "global", all IDs use the global minimum non-missing time.
input_units	Character string or NULL. Units of numeric input time. If NULL, numeric time is assumed to already be in the desired output scale and only subtraction is performed.
scale	Character string. Method used to compute the time-scaling divisor. If scale = "mean_dt", the divisor is the mean positive consecutive elapsed-time interval. If scale = "median_dt", the divisor is the median positive consecutive elapsed-time interval.
scale_value	Numeric or NULL. Optional user-supplied positive scaling divisor in the elapsed-time units. If supplied, this value is used instead of estimating the divisor from the data.
replace	Logical. If TRUE, replace the original time variable. If FALSE, append output.

Details

This scaling can improve numerical optimization because the drift and diffusion parameters are estimated with respect to a better-scaled time variable. However, the resulting model parameters are interpreted per CT-scaled time unit, not per original time unit.

If the original elapsed time is measured in the requested units and the scaling divisor is c , then

$$t_{\text{ct}} = t_{\text{original}}/c.$$

Consequently, drift and diffusion covariance parameters estimated using t_{ct} are on the CT-scaled time scale:

$$\Phi_{\text{ct}} = c\Phi_{\text{original}},$$

and, for a diffusion covariance or intensity matrix,

$$\Sigma_{\text{ct}} = c\Sigma_{\text{original}}.$$

To express estimates back in the original time units, divide drift and diffusion covariance/intensity estimates by the stored `scale_value`. If the diffusion parameter is a standard deviation or Cholesky factor rather than a covariance/intensity matrix, divide by `sqrt(scale_value)`.

The minimum positive consecutive interval is reported as a diagnostic but is not used as a scaling option. The goal is to make the typical consecutive interval approximately 1, not the smallest interval.

Value

Returns a data frame. The returned data frame has an attribute named "time_ct_scale" containing the scaling information, time-scale interpretation, and conversion multipliers for drift and diffusion parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2),
  datetime = as.POSIXct(
    c(
      "2020-01-01 00:00:00",
      "2020-01-01 06:00:00",
      "2020-01-03 12:00:00",
      "2020-01-04 00:00:00"
    ),
    tz = "UTC"
  )
)

data_ct <- ElapsedTimeByIDCT(
  data = data,
  id = "id",
  time = "datetime",
  output = "time_ct",
  units = "hours",
  scale = "mean_dt"
)

attr(data_ct, "time_ct_scale")
```

Description

The function removes IDs that do not satisfy simple minimum data requirements. For diagnostic screening and sensitivity-analysis decisions, prefer [DiagnosticsByID\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), and [ScreenByID\(\)](#).

Usage

```
FilterByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  min_rows = 2L,
  min_complete = 2L,
  max_prop_missing = 1,
  allow_initial_na = TRUE,
  allow_all_missing = FALSE
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
min_rows	Non-negative integer. Minimum number of rows required per ID.
min_complete	Non-negative integer. Minimum number of complete observed rows required per ID.
max_prop_missing	Numeric. Maximum allowed proportion of missing values across observed variables.

`allow_initial_na`
 Logical. If FALSE, remove IDs where the initial row contains missing observed values. Covariates are not used to determine whether the initial row has missing values.

`allow_all_missing`
 Logical. If FALSE, remove IDs where all observed values are missing.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:3, each = 3),
  time = rep(1:3, times = 3),
  y = c(1, 2, 3, NA, NA, 4, NA, NA, NA)
)
data

FilterByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  min_complete = 2
)
```

FilterObservedRows	<i>Remove Rows With Too Few Observed Values</i>
--------------------	---

Description

The function removes rows with fewer than a requested number of non-missing observed variables. This is useful after regularizing or padding intensive longitudinal data, where many rows may contain no observed measurements but retain an ID and time value.

Usage

```
FilterObservedRows(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  min_nonmissing = 1L,
  min_rows_by_id = NULL
)
```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.
<code>min_nonmissing</code>	Positive integer. Minimum number of non-missing observed variables required to retain a row.
<code>min_rows_by_id</code>	NULL or positive integer. If not NULL, IDs with fewer than <code>min_rows_by_id</code> retained rows are removed after row filtering.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2),
  time = c(1, 2, 3, 1, 2),
  y1 = c(1, NA, 3, NA, 2),
  y2 = c(NA, NA, 4, NA, 3)
)

FilterObservedRows(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

FlagDiagnosticsByID *Flag Potentially Problematic IDs*

Description

The function adds rule-based flags to the output of [DiagnosticsByID\(\)](#). It is intended for sensitivity checks before fitting dynamic models. It flags low observation counts, missingness, duplicate ID-time rows, non-finite observed values, large time gaps, low variability, and extreme observed values.

Usage

```
FlagDiagnosticsByID(
  x,
  min_observed_rows = 30L,
  min_complete_rows = NULL,
  max_prop_all_missing = 0.95,
  max_gap = NULL,
  max_median_gap = NULL,
  min_sd = 0.05,
  extreme_cut = 6,
  drop_score_cut = 2L
)
```

Arguments

x Data frame returned by [DiagnosticsByID\(\)](#).

min_observed_rows Positive integer. Minimum number of observed rows expected per ID.

min_complete_rows NULL or positive integer. Minimum number of complete rows expected per ID.

max_prop_all_missing	NULL or number between 0 and 1. Maximum tolerated proportion of all-missing observed rows.
max_gap	NULL or positive number. Maximum tolerated observed-row time gap in the units used by DiagnosticsByID() .
max_median_gap	NULL or positive number. Maximum tolerated median observed-row time gap in the units used by DiagnosticsByID() .
min_sd	NULL or non-negative number. Minimum tolerated within-ID standard deviation across observed variables. If NULL, low-SD flagging is skipped.
extreme_cut	Positive number. Absolute-value cutoff used to flag extreme observations.
drop_score_cut	Positive integer. IDs with priority scores greater than or equal to this value are marked as sensitivity-drop candidates.

Value

Returns x with flag columns appended.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

diagnostics <- DiagnosticsByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)

FlagDiagnosticsByID(
  x = diagnostics,
  min_observed_rows = 3,
  min_sd = 0.05
```

)

GetDropID

Get Sensitivity-Drop IDs From Diagnostics

Description

The function extracts IDs marked as sensitivity-drop candidates by [FlagDiagnosticsByID\(\)](#).

Usage

```
GetDropID(x, flagged_only = TRUE)
```

Arguments

x Data frame returned by [FlagDiagnosticsByID\(\)](#).

flagged_only Logical. If TRUE, return IDs marked as sensitivity-drop candidates. If FALSE, return all flagged IDs.

Value

Returns a vector of IDs.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

diagnostics <- DiagnosticsByID(
  data = data,
  id = "id",
```

```
time = "time",
observed = c("y1", "y2")
)

flagged <- FlagDiagnosticsByID(
  x = diagnostics,
  min_observed_rows = 3,
  min_sd = 0.05
)

GetDropID(flagged)
```

GetObservedInitial	<i>Get Observed Initial Means and Covariances</i>
--------------------	---

Description

The function extracts the first observed time point for each ID and computes the sample mean vector and sample covariance matrix of the observed variables across IDs. The resulting mean vector and covariance matrix can be used as fixed initial conditions in dynamic models.

Usage

```
GetObservedInitial(data, id, time, observed, ridge = 1e-06)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
ridge	Positive numeric. Small value added to the diagonal of the covariance matrix if the estimated covariance matrix is not positive definite.

Details

Rows are sorted by ID and time before selecting the first row for each ID. Missing observed values are ignored when computing the mean vector. The covariance matrix is computed using complete cases of the selected first-time observations.

If the covariance matrix is not positive definite, a small ridge value is added to the diagonal.

Value

Returns a list with the following elements:

- `data` Data frame containing the first observed time point for each ID and the selected observed variables.
- `mean` Numeric vector of observed-variable means computed from the first observed time point across IDs.
- `cov` Observed-variable covariance matrix computed from complete cases of the first observed time point across IDs.
- `n` Number of IDs with a first observed row.
- `n_complete` Number of IDs with complete data on all selected observed variables at the first observed time point.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6),
  y2 = c(4, 3, 2, 1, 5, 4, 3, 2, 6, 5, 4, 3)
)
data

init <- GetObservedInitial(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)

init$data
init$mean
init$cov
init$n
init$n_complete
```

InitialNA

*Check for Missing Observed Values in Initial Row by ID***Description**

The function checks whether the initial row for each ID has missing values in the observed variables. Covariates are retained when selecting and sorting the data, but they are not used to determine whether the initial row is missing.

Usage

```
InitialNA(data, id, time, observed, covariates = NULL)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a vector of ID values where the initial row has any missing observed value.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, 10, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, 10, 11, 12, 13, 14),
  y3 = c(1, NA, 3, 4, 5, 10, 11, 12, 13, 14)
)
data

InitialNA(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3)
)
```

InsertNA	<i>Insert NAs for Missing Observations</i>
----------	--

Description

The function creates a sequence of time values. This is a simple helper retained for compatibility. For new workflows, prefer [RegularizeTimeByID\(\)](#), which provides explicit global/by-ID grids and preserve/snap behavior.

Usage

```
InsertNA(data, id, time, observed, covariates = NULL, delta_t)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
delta_t	Positive number. Time interval.

Details

It starts with the smallest time value as the starting point and the largest time value as the endpoint. The sequence is incremented by `delta_t`. This new sequence is combined with the existing empirical time values. For any specific time value where there are no observations, NAs are inserted.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 4, 1, 3, 4),
  y1 = c(10, 11, 13, 20, 22, 23),
  y2 = c(5, 6, 8, 15, 17, 18)
)
data

InsertNA(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  delta_t = 1
)
```

Description

The function creates lagged observed variables within ID. Lagged values never cross ID boundaries.

Usage

```
LagByID(data, id, time, observed, covariates = NULL, lags = 1L, prefix = "lag")
```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.
<code>lags</code>	Positive integer vector. Lags to create.
<code>prefix</code>	Character string. Prefix for the lagged variable names.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = rnorm(6),
  y2 = rnorm(6)
)
data

LagByID(
```



```

    data = data,
    id = "id",
    time = "time",
    observed = c("y1", "y2"),
    lags = 1
  )

```

MakeClockTime

Make Clock Time

Description

The function combines a date and a clock time into a POSIXct vector. Clock time can be supplied as decimal hours, for example 13.5, or as a character time, for example "13:30" or "13:30:00".

Usage

```

MakeClockTime(
  date,
  time,
  tz = "UTC",
  date_formats = c("%m/%d/%y", "%m/%d/%Y", "%Y-%m-%d"),
  invalid = c("NA", "error")
)

```

Arguments

date	Date vector. Dates can be supplied as Date, POSIXt, or character values.
time	Numeric or character vector. Clock time in decimal hours, "HH:MM", or "HH:MM:SS" format.
tz	Character string. Time zone passed to as.POSIXct().
date_formats	Character vector. Date formats used to parse character dates.
invalid	Character string. If invalid = "NA", invalid clock times are returned as NA. If invalid = "error", invalid clock times throw an error.

Value

Returns a POSIXct vector.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
MakeClockTime(
  date = "2020-01-02",
  time = 13.5,
  tz = "America/New_York"
)
```

```
MakeClockTime(
  date = "01/02/20",
  time = "13:30",
  tz = "America/New_York"
)
```

PlotByID

Plot Observed Variables by ID

Description

The function creates one time-series plot for each observed variable. Within each plot, trajectories are overlaid by ID. Non-finite observed values, such as NaN, Inf, and -Inf, are treated as missing for plotting.

Usage

```
PlotByID(
  data,
  id,
  time,
  observed,
  ids = NULL,
  times = NULL,
  type = "b",
  pch = NULL,
  lty = NULL,
  col = NULL,
  xlab = NULL,
  ylab = NULL,
```

```

    main = NULL,
    xlim = NULL,
    ylim = NULL,
    legend = FALSE,
    ask = NULL,
    ...
)

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>ids</code>	Optional vector. ID values to plot. If NULL, all available IDs are plotted.
<code>times</code>	Optional vector of time values. If NULL, all available time points are plotted. If supplied, the range of times is used to subset the data. For example, <code>times = c(0, 9)</code> plots all rows with time values from 0 to 9, inclusive.
<code>type</code>	Character string. Plot type passed to <code>graphics::lines()</code> . Default is "b".
<code>pch</code>	Optional vector of plotting characters. If NULL, plotting characters are generated automatically by ID. Recycled to match the number of plotted IDs.
<code>lty</code>	Optional vector of line types. If NULL, line types are generated automatically by ID. Recycled to match the number of plotted IDs.
<code>col</code>	Optional vector of colors. If NULL, colors are generated automatically by ID. Recycled to match the number of plotted IDs.
<code>xlab</code>	Optional character string. Label for the x-axis. If NULL, the name of the time variable is used.
<code>ylab</code>	Optional character string. Label for the y-axis. If NULL, the observed variable name is used.
<code>main</code>	Optional character string. Plot title prefix. If NULL, the observed variable name is used.
<code>xlim</code>	Optional vector of length 2. Limits for the x-axis.
<code>ylim</code>	Optional numeric vector of length 2. Limits for the y-axis. If NULL, limits are computed separately for each observed variable using finite observed values only.
<code>legend</code>	Logical. If TRUE, add a legend identifying IDs.
<code>ask</code>	Logical. If TRUE, ask before drawing the next observed-variable plot. The default is TRUE in interactive sessions when plotting more than one observed variable.
<code>...</code>	Additional arguments passed to <code>graphics::lines()</code> .

Value

Invisibly returns the IDs that were plotted.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:4, each = 5),
  time = rep(1:5, times = 4),
  y1 = c(
    1, 2, 3, 4, 5,
    2, 3, 4, 5, 6,
    3, 4, 5, 6, 7,
    4, 5, 6, 7, 8
  ),
  y2 = c(
    5, 4, 3, 2, 1,
    6, 5, 4, 3, 2,
    7, 6, 5, 4, 3,
    8, 7, 6, 5, 4
  ),
  y3 = c(
    1, 1, 2, 2, 3,
    2, 2, 3, 3, 4,
    3, 3, 4, 4, 5,
    4, 4, 5, 5, 6
  )
)
data

PlotByID(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3),
  ask = FALSE
)
```

```

PlotByID(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3),
  ids = 1:3,
  times = c(1, 3),
  legend = TRUE,
  ask = FALSE
)

```

RegularizeTimeByID	<i>Regularize Time by ID</i>
--------------------	------------------------------

Description

The function inserts rows with missing values so that time follows a regular grid. A global grid uses the same time sequence for every ID. An ID-specific grid uses each ID's own minimum and maximum observed time values. If `method = "preserve"`, the function completes the regular grid while retaining empirical off-grid time values. If `method = "snap"`, empirical time values are snapped to the nearest grid point and the returned data use only grid time values.

Usage

```

RegularizeTimeByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  delta_t,
  grid = c("global", "by_id"),
  method = c("preserve", "snap")
)

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.

covariates	Character vector. A vector of character strings of the names of the covariates in the data.
delta_t	Positive number. Time interval.
grid	Character string. If grid = "global", use one time grid from the global minimum to the global maximum time value. If grid = "by_id", use an ID-specific time grid from each ID's minimum to maximum time value.
method	Character string. If method = "preserve", complete the regular grid but preserve observed off-grid time values. If method = "snap", snap observed time values to the nearest grid point so the output contains only regular grid times. When multiple rows for the same ID snap to the same grid point, the row closest to the grid point is kept; ties are broken by the largest number of non-missing observed/covariate values and then by original order.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2),
  time = c(1, 3, 1, 2),
  y = c(10, 12, 20, 21)
)
data

RegularizeTimeByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  delta_t = 1,
  grid = "by_id"
)

RegularizeTimeByID(
```

```

data = data.frame(
  id = c(1, 1),
  time = c(1.0, 1.7),
  y = c(10, 17)
),
id = "id",
time = "time",
observed = "y",
delta_t = 0.5,
grid = "by_id",
method = "snap"
)

```

ReplaceMissingCode	<i>Replace Missing-Value Codes</i>
--------------------	------------------------------------

Description

The function replaces user-specified missing-value codes with NA in a data frame. The replacement is applied column by column and preserves the original column classes whenever possible.

Usage

```
ReplaceMissingCode(data, values = c(-999, "-999"), columns = names(data))
```

Arguments

data	Data frame.
values	Vector. Values to replace with NA.
columns	Character vector. Names of the columns where replacement should be applied. If NULL, no columns are modified.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = 1:3,
  y = c(1, -999, 3),
  x = c("a", "-999", "c"),
  stringsAsFactors = FALSE
)
data

ReplaceMissingCode(
  data = data,
  values = c(-999, "-999")
)
```

ResolveCloseTimeByID *Resolve Close Time Points by ID*

Description

The function resolves observations that occur very close together within ID. Within each ID, rows are ordered by time and grouped into close-time clusters. Consecutive observations belong to the same cluster when the time gap between them is less than `min_gap`. One row is retained from each close-time cluster.

Usage

```
ResolveCloseTimeByID(
  data,
  id,
  time,
  observed,
  min_gap,
  method = c("max_complete", "first", "last")
)
```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.

min_gap	Numeric scalar or difftime. Minimum allowed gap between consecutive observations within ID. For numeric time, min_gap must be a positive finite numeric scalar in the same units as time. For Date or POSIXt time, min_gap must be a positive difftime object.
method	Character string. Method used to select the row retained from each close-time cluster. If method = "max_complete", the row with the largest number of non-missing observed variables is retained. Ties are resolved by retaining the earliest row in the sorted data. If method = "first", the first row in the cluster is retained. If method = "last", the last row in the cluster is retained.

Details

This is useful before fitting continuous-time models, where observations that are nearly simultaneous can create numerical problems, especially when the observed values change substantially over a very small time interval.

Value

Returns a data frame with close-time observations resolved within ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 1, 2, 2),
  datetime = as.POSIXct(
    c(
      "2020-01-01 00:00:00",
      "2020-01-01 00:02:00",
      "2020-01-01 01:00:00",
      "2020-01-01 01:03:00",
      "2020-01-01 00:00:00",
      "2020-01-01 00:10:00"
    )
  ),
  tz = "UTC"
),
y1 = c(1, NA, 3, 4, 5, 6),
```

```

    y2 = c(NA, 2, 3, 4, 5, 6)
  )

  ResolveCloseTimeByID(
    data = data,
    id = "id",
    time = "datetime",
    observed = c("y1", "y2"),
    min_gap = as.difftime(
      5,
      units = "mins"
    ),
    method = "max_complete"
  )

```

ResolveDuplicateIDTime

Resolve Duplicate ID-Time Rows

Description

The function resolves exact duplicate id-time rows using a simple deterministic rule. This is useful before inserting missing rows on a time grid because grid-completion functions generally require unique id-time combinations.

Usage

```

ResolveDuplicateIDTime(
  data,
  id,
  time,
  observed = NULL,
  covariates = NULL,
  method = c("first", "last", "max_complete"),
  order_by = NULL,
  decreasing = FALSE
)

```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.

observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
method	Character string. Rule used to choose one row from each duplicated id-time group. If method = "first", keep the first row after sorting by id and time. If method = "last", keep the last row after sorting by id and time. If method = "max_complete", keep the row with the largest number of non-missing values in observed.
order_by	Character vector. Optional columns used to break ties after the main method rule.
decreasing	Logical. If TRUE, sort order_by columns in decreasing order.

Value

Returns a data frame with unique id-time rows.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1),
  time = c(0, 0, 1),
  y1 = c(NA, 1, 2),
  y2 = c(NA, 1, 3)
)
data

ResolveDuplicateIDTime(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  method = "max_complete"
)
```

RoundClockTime

*Round Clock Time***Description**

The function rounds, floors, or ceilings clock times to a specified unit. It supports POSIXt, Date, and numeric time values.

Usage

```
RoundClockTime(
  x,
  unit = "hour",
  method = c("round", "floor", "ceiling"),
  origin = "1970-01-01",
  tz = "UTC"
)
```

Arguments

x	A POSIXt, Date, or numeric vector.
unit	Character string. Unit to round to. Supported values are "second", "minute", "hour", "day", and their plural forms.
method	Character string. One of "round", "floor", or "ceiling".
origin	Origin for numeric time values. Only used if x is numeric.
tz	Character string. Time zone used when numeric time values are converted to POSIXct.

Value

Returns a vector with rounded times.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
x <- as.POSIXct(
  "2020-01-01 10:31:00",
  tz = "America/New_York"
)

RoundClockTime(
  x = x,
  unit = "hour"
)
```

ScaleByID

*Scale by ID***Description**

The function scales the data by ID.

Usage

```
ScaleByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  scale = TRUE,
  obs_skip = NULL,
  cov_skip = NULL
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

scale	Logical. If scale = TRUE, standardize by id. If scale = FALSE, mean center by id.
obs_skip	Character vector. A vector of character strings of the names of the observed variables to skip centering/scaling.
cov_skip	Character vector. A vector of character strings of the names of the covariates to skip centering/scaling.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(
    1, 2, 3, 4,
    10, 11, 12, 13,
    20, 22, 24, 26
  ),
  y2 = c(
    4, 3, 2, 1,
    13, 12, 11, 10,
    26, 24, 22, 20
  )
)
data

ScaleByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)

ScaleByID(
```

```

    data = data,
    id = "id",
    time = "time",
    observed = c("y1", "y2"),
    scale = FALSE
  )

```

ScreenByID

Screen IDs Using Dynamic-Data Diagnostics

Description

The function computes ID-level diagnostics, applies rule-based flags, and returns IDs that are candidates for exclusion or sensitivity analysis. It is a wrapper around [DiagnosticsByID\(\)](#), [FlagDiagnosticsByID\(\)](#), and [GetDropID\(\)](#).

Usage

```

ScreenByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  missing_codes = NULL,
  min_nonmissing = 1L,
  extreme_cut = c(4, 5, 6),
  sd_cut = c(0.1, 0.05),
  time_scale = 1,
  posix_unit = "hours",
  min_observed_rows = 30L,
  min_complete_rows = NULL,
  max_prop_all_missing = 0.95,
  max_gap = NULL,
  max_median_gap = NULL,
  min_sd = 0.05,
  flag_extreme_cut = 6,
  drop_score_cut = 2L,
  flagged_only = TRUE
)

```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
------	---

id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
missing_codes	NULL or vector. Values to temporarily treat as missing when computing diagnostics. The input data are not modified.
min_nonmissing	Positive integer. Minimum number of finite observed variables required for a row to count as an observed row.
extreme_cut	Numeric vector. Absolute-value cutoffs used to count extreme values.
sd_cut	Numeric vector. Within-ID standard-deviation cutoffs used to count low-variance variables.
time_scale	Positive number. Multiplier applied to numeric time gaps. For example, use <code>time_scale = 24</code> when time is measured in days and gaps should be reported in hours.
posix_unit	Character string. Unit for gaps when time is a POSIXt or Date variable. Passed to <code>as.numeric.difftime()</code> .
min_observed_rows	Positive integer. Minimum number of observed rows expected per ID.
min_complete_rows	NULL or positive integer. Minimum number of complete rows expected per ID.
max_prop_all_missing	NULL or number between 0 and 1. Maximum tolerated proportion of all-missing observed rows.
max_gap	NULL or positive number. Maximum tolerated observed-row time gap in the units used by DiagnosticsByID() .
max_median_gap	NULL or positive number. Maximum tolerated median observed-row time gap in the units used by DiagnosticsByID() .
min_sd	NULL or non-negative number. Minimum tolerated within-ID standard deviation across observed variables. If NULL, low-SD flagging is skipped.
flag_extreme_cut	Positive number. Absolute-value cutoff used by FlagDiagnosticsByID() . This value is added to <code>extreme_cut</code> before calling DiagnosticsByID() so that the required total-count column is available.
drop_score_cut	Positive integer. IDs with priority scores greater than or equal to this value are marked as sensitivity-drop candidates.
flagged_only	Logical. If TRUE, return IDs marked as sensitivity-drop candidates. If FALSE, return all flagged IDs.

Value

Returns a list with the following elements:

`diagnostics` Diagnostics with flag columns appended.

`drop_id` IDs selected by [GetDropID\(\)](#).

`keep_id` IDs not in `drop_id`.

`call` The matched call.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

ScreenByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  min_observed_rows = 3,
  min_sd = 0.05
)
```

SubsetByID

Subset Data Set by ID

Description

The function creates a list of data frames for each ID.

Usage

```
SubsetByID(data, id, time, observed, covariates = NULL)
```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a list by ID numbers.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(
    1, 2, 3, 4,
    10, 11, 12, 13,
    20, 21, 22, 23
  ),
  y2 = c(
    4, 3, 2, 1,
    13, 12, 11, 10,
    23, 22, 21, 20
  )
)
```

```
    )
  )
data

SubsetByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

SummaryByID	<i>Summarize Dynamic Modeling Data by ID</i>
-------------	--

Description

The function returns a simple diagnostic table with one row per ID. For model-screening workflows, prefer [DiagnosticsByID\(\)](#), [FlagDiagnosticsByID\(\)](#), and [ScreenByID\(\)](#).

Usage

SummaryByID(data, id, time, observed, covariates = NULL)

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

Details

The `initial_na` column is based on the observed variables only. Covariates are retained in the input checks and sorted data, but missing covariates in the first row do not cause `initial_na = TRUE`.

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = c(1, NA, 3, 4, 5, 6),
  y2 = c(1, 2, 3, NA, NA, NA)
)
data

SummaryByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

TrimInitialRowsByID *Trim Leading Rows With Too Few Observed Values by ID*

Description

The function removes leading rows within each ID until the first retained row has at least `min_nonmissing` non-missing observed variables. This is useful before creating an elapsed-time variable when leading records contain no usable measurements. Unlike [DeleteInitialNA\(\)](#), the default does not require the first retained row to be complete on all selected variables.

Usage

```
TrimInitialRowsByID(
  data,
  id,
  time,
  observed,
```

```

    covariates = NULL,
    min_nonmissing = 1L
  )

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.
<code>min_nonmissing</code>	Positive integer. Minimum number of non-missing observed variables required in the first retained row for each ID.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnoseScaleByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [ElapsedTimeByIDCT\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveCloseTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#)

Examples

```

data <- data.frame(
  id = rep(1:2, each = 4),
  time = rep(1:4, times = 2),
  y1 = c(NA, 1, 2, 3, NA, NA, 1, 2),
  y2 = c(NA, NA, 2, 3, NA, 1, 1, 2)
)

TrimInitialRowsByID(
  data = data,

```

```
id = "id",  
time = "time",  
observed = c("y1", "y2"),  
min_nonmissing = 1  
)
```

Index

* **Dynamic Modeling Utility Functions**

- CheckDynData, [2](#)
- CombineByIDTime, [4](#)
- DeleteInitialNA, [5](#)
- DeleteObservedAllNA, [7](#)
- DeltaTByID, [8](#)
- DetrendByID, [10](#)
- DiagnoseScaleByID, [12](#)
- DiagnosticsByID, [13](#)
- DropByID, [15](#)
- ElapsedTimeByID, [16](#)
- ElapsedTimeByIDCT, [18](#)
- FilterByID, [21](#)
- FilterObservedRows, [22](#)
- FlagDiagnosticsByID, [24](#)
- GetDropID, [26](#)
- GetObservedInitial, [27](#)
- InitialNA, [29](#)
- InsertNA, [30](#)
- LagByID, [31](#)
- MakeClockTime, [33](#)
- PlotByID, [34](#)
- RegularizeTimeByID, [37](#)
- ReplaceMissingCode, [39](#)
- ResolveCloseTimeByID, [40](#)
- ResolveDuplicateIDTime, [42](#)
- RoundClockTime, [44](#)
- ScaleByID, [45](#)
- ScreenByID, [47](#)
- SubsetByID, [49](#)
- SummaryByID, [51](#)
- TrimInitialRowsByID, [52](#)

* **data**

- CheckDynData, [2](#)
- CombineByIDTime, [4](#)
- DeleteInitialNA, [5](#)
- DeleteObservedAllNA, [7](#)
- DeltaTByID, [8](#)
- DetrendByID, [10](#)

- DiagnoseScaleByID, [12](#)
- DiagnosticsByID, [13](#)
- DropByID, [15](#)
- ElapsedTimeByID, [16](#)
- ElapsedTimeByIDCT, [18](#)
- FilterByID, [21](#)
- FilterObservedRows, [22](#)
- FlagDiagnosticsByID, [24](#)
- GetDropID, [26](#)
- GetObservedInitial, [27](#)
- InitialNA, [29](#)
- InsertNA, [30](#)
- LagByID, [31](#)
- MakeClockTime, [33](#)
- RegularizeTimeByID, [37](#)
- ReplaceMissingCode, [39](#)
- ResolveCloseTimeByID, [40](#)
- ResolveDuplicateIDTime, [42](#)
- RoundClockTime, [44](#)
- ScaleByID, [45](#)
- ScreenByID, [47](#)
- SubsetByID, [49](#)
- SummaryByID, [51](#)
- TrimInitialRowsByID, [52](#)

* **dynTools**

- CheckDynData, [2](#)
- CombineByIDTime, [4](#)
- DeleteInitialNA, [5](#)
- DeleteObservedAllNA, [7](#)
- DeltaTByID, [8](#)
- DetrendByID, [10](#)
- DiagnoseScaleByID, [12](#)
- DiagnosticsByID, [13](#)
- DropByID, [15](#)
- ElapsedTimeByID, [16](#)
- ElapsedTimeByIDCT, [18](#)
- FilterByID, [21](#)
- FilterObservedRows, [22](#)
- FlagDiagnosticsByID, [24](#)

- GetDropID, 26
- GetObservedInitial, 27
- InitialNA, 29
- InsertNA, 30
- LagByID, 31
- MakeClockTime, 33
- PlotByID, 34
- RegularizeTimeByID, 37
- ReplaceMissingCode, 39
- ResolveCloseTimeByID, 40
- ResolveDuplicateIDTime, 42
- RoundClockTime, 44
- ScaleByID, 45
- ScreenByID, 47
- SubsetByID, 49
- SummaryByID, 51
- TrimInitialRowsByID, 52
- * plot**
 - PlotByID, 34
- CheckDynData, 2
- CheckDynData(), 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- CombineByIDTime, 4
- CombineByIDTime(), 3, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- DeleteInitialNA, 5
- DeleteInitialNA(), 3, 5, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- DeleteObservedAllNA, 7
- DeleteObservedAllNA(), 3, 5, 6, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- DeltaTByID, 8
- DeltaTByID(), 3, 5, 6, 8, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- DetrendByID, 10
- DetrendByID(), 3, 5, 6, 8, 9, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- DiagnoseScaleByID, 12
- DiagnoseScaleByID(), 3, 5, 6, 8, 9, 11, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- DiagnosticsByID, 13
- DiagnosticsByID(), 3, 5, 6, 8, 9, 11, 13, 16, 17, 20–26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46–53
- DropByID, 15
- DropByID(), 3, 5, 6, 8, 9, 11, 13, 15, 17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- ElapsedTimeByID, 16
- ElapsedTimeByID(), 3, 5, 6, 8, 9, 11, 13, 15, 16, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- ElapsedTimeByIDCT, 18
- ElapsedTimeByIDCT(), 3, 5, 6, 8, 9, 11, 13, 15–17, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- FilterByID, 21
- FilterByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- FilterObservedRows, 22
- FilterObservedRows(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- FlagDiagnosticsByID, 24
- FlagDiagnosticsByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20–23, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46–53
- GetDropID, 26
- GetDropID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20–23, 25, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 47, 49, 50, 52, 53
- GetObservedInitial, 27
- GetObservedInitial(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- graphics::lines(), 35
- InitialNA, 29

- InitialNA(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- InsertNA, 30
- InsertNA(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- LagByID, 31
- LagByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- MakeClockTime, 33
- MakeClockTime(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- PlotByID, 34
- PlotByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 38, 39, 41, 43, 44, 46, 49, 50, 52, 53
- RegularizeTimeByID, 37
- RegularizeTimeByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28–32, 34, 36, 39, 41, 43, 44, 46, 49, 50, 52, 53
- ReplaceMissingCode, 39
- ReplaceMissingCode(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 41, 43, 44, 46, 49, 50, 52, 53
- ResolveCloseTimeByID, 40
- ResolveCloseTimeByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 43, 44, 46, 49, 50, 52, 53
- ResolveDuplicateIDTime, 42
- ResolveDuplicateIDTime(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 44, 46, 49, 50, 52, 53
- RoundClockTime, 44
- RoundClockTime(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 46, 49, 50, 52, 53
- ScaleByID, 45
- ScaleByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 49, 50, 52, 53
- ScreenByID, 47
- ScreenByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20–23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 50–53
- SubsetByID, 49
- SubsetByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 52, 53
- SummaryByID, 51
- SummaryByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 53
- TrimInitialRowsByID, 52
- TrimInitialRowsByID(), 3, 5, 6, 8, 9, 11, 13, 15–17, 20, 22, 23, 25, 26, 28, 29, 31, 32, 34, 36, 38, 39, 41, 43, 44, 46, 49, 50, 52